NN
ACADEMY

IGCSE Maths Paper 4 Worked Solutions

Cambridge IGCSE Mathematics 0580, Paper 4 Extended — calculator allowed. Full worked solutions to all twelve questions of the NurNabi Academy Full Revision Paper, solved the simplest way, with the method and accuracy marks shown at every step, the way a Cambridge examiner awards them.

12	130	Q1–Q12	6
questions	marks	E1.11 – E9.7	videos · 2h 15m

M1 method mark

A1 accuracy mark (needs the M mark)

B1 independent mark

HOW THESE SOLUTIONS ARE MARKED

Reading the mark scheme

What the three letters mean

- **M1 — method.** Awarded for a correct method, even if the arithmetic that follows is wrong. This is the mark students throw away by writing only the answer.
- **A1 — accuracy.** The correct value, and it is **dependent** on the M mark. A right answer with no method shown can still score it, but a wrong method never does.
- **B1 — independent.** Awarded for the answer on its own, with no method required. Usually short, one-step parts.
- **FT — follow through.** If an earlier part went wrong, later parts are marked on the student's own (wrong) value. This is why you never stop after one mistake.

Say this while you record

- Name each mark out loud as you earn it: "that is M1 for the cosine rule".
- Write the unrounded value, then round once. Say why.

The accuracy rule for Paper 4. Unless a question says otherwise, give answers correct to **3 significant figures**, and angles correct to **1 decimal place**. Exact answers — fractions, surds, multiples of π — are always accepted in place of a decimal.

The single most expensive habit. Rounding part way through. If you write 122.6 and then use 122.6 in the next part, your final answer can fall outside the accepted range and lose the A mark even though your method was perfect. Keep the full value in the calculator, and round only the answer you write down.

Show that… questions. The answer is printed in the question, so you earn nothing for writing it. The marks are entirely for the working that leads to it — and you must reach **more** accuracy than the printed value before you round to it.

Give a reason… questions. The angle alone scores nothing without the theorem named in full. "Angles in a semicircle" earns the mark; "because it is 90°" does not.

Question 1

Profit, ratio and interest

A shop buys a laptop for $540 and sells it at a profit of 35%. Calculate the selling price. [2]

Multiplier: a 35% profit means 100% + 35% = **135%** = 1.35 M1

540 × 1.35

$729 A1

In a sale, the price of a printer is reduced by 18% to $287. Calculate the original price. [2]

Reverse percentage. $287 is not the original, it is 82% of it.

Original × 0.82 = 287, so original = 287 ÷ 0.82 M1

$350 A1

Marks: M1 for 540 × 1.35 or for 540 × 0.35 = 189 added on. A1 for 729.

The two directions. Going *forward* you multiply: 540 × 1.35. Coming *back* you divide: 287 ÷ 0.82. Ask yourself every time which number is the 100%.

Common mistake: finding 18% of 287 and adding it on, giving $338.66. That is 18% of the *sale* price, not of the original. Check: 350 × 0.82 = 287 ✓, but 338.66 × 0.82 = 277.70 ✗.

Check it: every reverse-percentage answer can be verified in one line. Multiply your answer forward and see if you land on the given number.

QUESTION 1 CONTINUED

Amira, Bilal and Chen share $4560 in the ratio 5 : 7 : 12. Calculate the amount each receives. [2]

Total parts: 5 + 7 + 12 = **24**

One part = 4560 ÷ 24 = **190** M1

5 × 190 = 950 7 × 190 = 1330 12 × 190 = 2280

Amira $950 · Bilal $1330 · Chen $2280 A1

Amira invests her share at 3.4% per year compound interest. Find the value after 6 years, to the nearest dollar. [3]

Compound interest is one multiplier, to a power.

950×1.034^6 M2

$= 950 \times 1.222146\ldots = 1161.039\ldots$

$1161 A1

Marks: M1 for 4560 ÷ 24, A1 for all three amounts. Then M2 for 950×1.034^6 (M1 only if the power or the multiplier is wrong), A1 for 1161.

Always find one part first. It makes every share a one-step multiplication and it gives you a check: 950 + 1330 + 2280 = 4560 ✓.

Common mistake: doing six separate years of interest on a rounded figure. You will drift by a dollar or two and lose the A mark. One multiplier, one power, one rounding — at the end.

Simple vs compound: simple interest is $P \times r \times n$ added on; compound is $P \times (\text{multiplier})^n$. Read which one the question asks for before you touch the calculator.

QUESTION 1 CONTINUED

Bilal invests his share ($1330) at r% per year **simple** interest. After 8 years it is worth $1968.40. Find r. [3]

Interest earned: 1968.40 − 1330 = **$638.40** M1

Simple interest is the same every year:

638.40 ÷ 8 = **$79.80** per year M1

As a percentage of the $1330 he started with:

$$\frac{79.80}{1330} \times 100$$

$r = 6$ A1

Marks: M1 for subtracting to get the interest, M1 for dividing by 8 (or for setting up 1330 × r × 8 ÷ 100 = 638.40), A1 for 6.

The formula way, if you prefer it: $\frac{1330 \times r \times 8}{100} = 638.40$ gives $106.4r = 638.40$, so $r = 6$. Same marks.

Common mistake: dividing 1968.40 by 1330 and treating the result as the rate. That gives the *total* growth factor 1.48, not the yearly rate. Always subtract the principal off first.

Check: 6% of 1330 = 79.80 a year; × 8 = 638.40; + 1330 = 1968.40 ✓.

Question 2

Algebraic manipulation

Solve $3x + 2y = 4$ and $5x - 4y = 25$. [3]

Elimination. The y terms are $+2$ and -4, so doubling the first equation makes them cancel.

$\times 2$: $6x + 4y = 8$ M1

Add: $11x = 33$, so $x = 3$ A1

Substitute into $3x + 2y = 4$:

$9 + 2y = 4 \rightarrow 2y = -5$

$$x = 3,\ y = -2.5$$ A1

Marks: M1 for a correct multiplication to match coefficients, A1 for one correct variable, A1 for the other.

Add or subtract? If the matched coefficients have *opposite* signs, add. If they have the *same* sign, subtract. Here $+4y$ and $-4y$ are opposite, so add.

Common mistake: doubling the left-hand side and forgetting the right. Everything gets multiplied, including the 4.

Always substitute back into the *other* equation to check: $5(3) - 4(-2.5) = 15 + 10 = 25$ ✓. Thirty seconds, and it catches a sign error every time.

QUESTION 2 CONTINUED

Simplify fully. $\dfrac{3x^2 - 12}{x^2 - x - 6}$ [3]

Factorise top and bottom separately.

Top: take out 3 first → $3(x^2 - 4) = 3(x - 2)(x + 2)$ B1

Bottom: two numbers multiplying to −6, adding to −1 → −3 and +2

$= (x - 3)(x + 2)$ B1

Cancel the common factor $(x + 2)$:

$$\frac{3(x-2)}{x-3}$$ B1

Marks: B1 for factorising the numerator, B1 for the denominator, B1 for cancelling to the final form.

Take the number out first. $3x^2 - 12$ does not look like a difference of two squares until you pull out the 3. Once you do, $x^2 - 4$ is obvious.

Never cancel across a + or − sign. You cannot cancel the x^2 terms, and you cannot cancel the 3 with the 6. Only whole bracketed factors cancel.

Why "fully" matters: leaving $\frac{3(x-2)(x+2)}{(x-3)(x+2)}$ scores two of the three marks. The last mark is for the cancelling.

QUESTION 2 CONTINUED

Make x the subject. $y = \dfrac{3x + 2}{x - 5}$ [4]

Clear the fraction first.

$y(x - 5) = 3x + 2$ M1

$xy - 5y = 3x + 2$ M1

Collect every x term on one side, everything else on the other:

$xy - 3x = 5y + 2$ M1

Factorise out x: $x(y - 3) = 5y + 2$

$$\boldsymbol{x = \dfrac{5y + 2}{y - 3}}$$ A1

Factorise fully. $6x^2 - 7x - 3$ [2]

$6 \times -3 = -18$. Two numbers: $\times$ to -18, $+$ to $-7 \rightarrow -9$ and $+2$ M1

$$\boldsymbol{(3x + 1)(2x - 3)}$$ A1

Marks: M1 multiply up, M1 expand, M1 collect the x terms and factorise, A1 for the final form.

The whole skill is step three. Whenever the new subject appears *twice*, you must gather both terms and take it out as a common factor. Nothing else will work.

Common mistake: writing $x = \dfrac{5y + 2}{y}$. You cannot divide by y alone — the x is inside a bracket with it.

For the factorising: split the middle term — $6x^2 - 9x + 2x - 3 = 3x(2x - 3) + 1(2x - 3)$. Expand your answer to check; it takes ten seconds.

Question 3

Graph of a cubic

Complete the table for $y = x^3 - 3x - 1$. [2]

$x = -1.5$: $(-1.5)^3 - 3(-1.5) - 1 = -3.375 + 4.5 - 1$ **B1**

$x = 1.5$: $(1.5)^3 - 3(1.5) - 1 = 3.375 - 4.5 - 1$ **B1**

0.125 and −2.125

Draw the graph for $-2 \leqslant x \leqslant 2.5$. [4]

Plot all ten points, then join them with one smooth curve. **B4**

The curve turns at about (−1, 1) and (1, −3) — if your sketch has no turning points you have joined the points in the wrong order.

Marks: B1 for each correct table value. Then B4 for the curve: B3 if up to two points are misplotted, B2 if the points are right but joined badly.

Where the marks go on a curve. A ruler between points, a line that stops short of $x = 2.5$, a thick feathered double line, or a flat bottom instead of a turning point — each of these loses a mark even when every plotted point is correct.

Negative cubes. $(-1.5)^3 = -3.375$, not +3.375. On the calculator put the bracket in. This single slip is the most common lost B mark in the whole question.

Sanity check the shape: a positive cubic comes up from the bottom left and leaves at the top right. Yours should too.

QUESTION 3 CONTINUED

Use your graph to solve $x^3 - 3x - 1 = 0$. [3]

Solving = 0 means: where does the curve cross the x-axis?

Read off all three crossings: B3

$$x = -1.53, -0.35, 1.88$$

By drawing a suitable tangent, estimate the gradient at $x = 2$. [3]

Draw a tangent touching the curve at (2, 1) — one straight line, resting on the curve, not cutting it. M1

Take two points far apart on **your tangent** and form the triangle: M1

gradient = $\frac{\text{rise}}{\text{run}}$

$$\approx 9$$ A1

Marks: B3 for all three roots (B2 for two, B1 for one), each within ±0.05. Then M1 for a tangent at $x = 2$, M1 for a correct rise ÷ run from it, A1 for a value in the accepted range.

The accepted range is wide. For the gradient, anything from about 7.5 to 10.5 scores, because it depends on your own tangent. The exact value is 9 (by calculus, $\frac{dy}{dx} = 3x^2 - 3 = 9$ at $x = 2$) — but you must still draw the tangent to earn the method marks.

Common mistake: quoting 9 from differentiation with no tangent drawn. The question says *by drawing a suitable tangent*, so a bare 9 scores **zero** for the method marks.

Make the triangle big. A rise-and-run over one small square magnifies every reading error. Use at least half the width of the grid.

Question 4

A composite solid

Show that the volume of the solid is 2149 cm^3, to 4 s.f. [2]

Cylinder: $\pi r^2 h = \pi \times 6^2 \times 15 = 540\pi$

Hemisphere (half a sphere): $\frac{1}{2} \times \frac{4}{3}\pi r^3 = \frac{2}{3}\pi \times 216 = 144\pi$ M1

Total = $540\pi + 144\pi = 684\pi = 2148.849\ldots$

= 2149 cm^3 (4 s.f.) A1

Calculate the total surface area. [4]

Curved cylinder: $2\pi rh = 2\pi(6)(15) = 180\pi$ M1

Base circle: $\pi r^2 = 36\pi$ M1

Curved hemisphere: $\frac{1}{2} \times 4\pi r^2 = 72\pi$ M1

Total = $288\pi = 904.778\ldots$

904.8 cm² (1 d.p.) · 905 cm² (3 s.f.) A1

Marks: M1 for the hemisphere as half a sphere, A1 for reaching 2148.8… *before* rounding. Then M1 for each of the three correct surfaces, A1 for 905.

The surface-area trap. There are **three** surfaces, not four. The flat circle at the *top* of the cylinder is hidden under the hemisphere, so it is not part of the surface. Counting it gives $324\pi = 1018$ and loses two marks.

Work in π all the way. $540\pi + 144\pi = 684\pi$ is exact and takes one line. Decimals at every stage invite rounding drift — fatal in a "show that".

"Show that" rule: you must display more accuracy than the printed answer. Writing "= 2149" straight away scores nothing; "= 2148.849… = 2149" scores the mark.

QUESTION 4 CONTINUED

The solid is melted into spheres of radius 1.6 cm. Find the greatest number of complete spheres. [3]

One sphere: $\frac{4}{3}\pi(1.6)^3 = 17.157\ldots$ cm^3 **M1**

Number = $\frac{2148.849\ldots}{17.157\ldots} = 125.24\ldots$ **M1**

Complete spheres — so round **down**, whatever the decimal.

125 spheres **A1**

A similar solid has surface area 2262 cm^2. Find its volume. [3]

Area scale factor = $\frac{2262}{904.78\ldots} = 2.5$ **M1**

Length SF = $\sqrt{2.5} = 1.5811\ldots$; Volume SF = $2.5^{1.5} = 3.9528\ldots$ **M1**

Volume = $2148.849\ldots \times 3.9528\ldots = 8494.4\ldots$

8490 cm^3 (3 s.f.) **A1**

Marks: M1 sphere volume, M1 divide, A1 for 125. Then M1 area SF, M1 convert to a volume SF, A1 for 8490.

The similar-solids ladder. If lengths scale by k, areas scale by k^2 and volumes by k^3. You are given an *area* ratio, so square-root it to get k, then cube it. Never go straight from area to volume.

Common mistake: multiplying the volume by 2.5, giving 5372. That treats an area ratio as a volume ratio. The correct factor is $2.5^{3/2} = 3.95$, not 2.5.

Round down, always. 125.24 spheres means 125 whole ones with material left over. "Greatest number of complete…" is never rounded to nearest.

Question 5

Cosine rule, sine rule and bearings

AB = 84 m, BC = 67 m, angle ABC = 108°. Calculate AC. [4]

Two sides and the angle **between** them — cosine rule. M1

$AC^2 = 84^2 + 67^2 - 2(84)(67)\cos 108°$ M1

$= 7056 + 4489 - 11256 \times (-0.30902...)$

$= 7056 + 4489 + 3478.3... = 15023.3...$ M1

AC = 122.6 m (1 d.p.) · 123 m (3 s.f.) A1

Marks: M1 for choosing the cosine rule, M1 for correct substitution, M1 for evaluating to 15023..., A1 for 122.6.

The obtuse angle is doing the work. cos 108° is **negative**, so the $-2bc\cos A$ term *adds* on. If your AC came out smaller than 84, you dropped the minus sign.

Check the calculator is in degrees. In radian mode cos 108 = −0.4755 and AC = 129.9 — a plausible-looking number that scores zero for accuracy. Look for **DEG** on the display before you start the paper.

Keep 122.5696... in the calculator — parts (b) and (e) both use it. Retyping 122.6 shifts the later answers.

QUESTION 5 CONTINUED

Calculate angle BAC. [3]

Now you have a side and its opposite angle — sine rule. M1

$$\frac{\sin BAC}{67} = \frac{\sin 108°}{122.5696...}$$ M1

$$\sin BAC = \frac{67 \times 0.95106...}{122.5696...} = 0.51987...$$

Angle BAC = 31.3° A1

The bearing of B from A is 052° and C lies east of AB. Find the bearing of C from A. [2]

A bearing is measured clockwise from north. B is at 052° from A, and C is a further 31.3° clockwise. M1

052 + 31.3 = 83.3

083.3° A1

Marks: M1 for the sine rule, M1 for correct rearrangement, A1 for 31.3°. Then M1 for 52 + their angle BAC, A1 for 083.3° (follow through from part b).

Write bearings with three figures. 83.3° should be written **083.3°**. Cambridge accepts both, but the three-figure habit stops you mis-reading your own work later.

Watch which angle is opposite which side. Angle BAC sits opposite side BC = 67, and the 108° sits opposite AC. Pairing them the wrong way is the usual error here.

Follow through saves you. Even if your AC was wrong in part (a), using it correctly here still earns both marks in (b) and (c).

QUESTION 5 CONTINUED

Calculate the area of triangle ABC. [2]

Two sides and the included angle — use $\frac{1}{2}ab\sin C$.

$\frac{1}{2} \times 84 \times 67 \times \sin 108°$ M1

$= 2814 \times 0.95106... = 2676.27...$

2680 m² (3 s.f.) A1

A vertical mast of height 30 m stands at C. Find the angle of elevation of its top from A. [2]

The mast is vertical, so this is a **right-angled** triangle: 30 m up, AC = 122.5696... m along.

$\tan\theta = \frac{30}{122.5696...} = 0.24476...$ M1

θ = 13.8° A1

Marks: M1 for ½(84)(67)sin 108°, A1 for 2680. Then M1 for tan θ = 30 ÷ their AC, A1 for 13.8°.

Angle of elevation is measured from the horizontal, looking up. The 30 m is the opposite side and AC is the adjacent, so it is tangent — no sine rule needed anywhere in this part.

Common mistake: using 84 or 67 as the horizontal distance. The mast is at C and you are looking from A, so the horizontal distance is **AC**, the side you worked out in part (a).

Area formula caution: $\frac{1}{2}ab\sin C$ needs the angle **between** the two sides you use. Angle B sits between AB and BC — which is exactly the pair given.

Question 6

Grouped data

Calculate an estimate of the mean time. [4]

Midpoints (you have no raw data, so assume each value sits at the centre of its class): M1

10	25	32.5	37.5	50

Σfx: 8(10) + 22(25) + 30(32.5) + 34(37.5) + 26(50) M1

= 80 + 550 + 975 + 1275 + 1300 = **4180** M1

Mean = $\frac{4180}{120}$ = 34.833…

34.8 minutes (3 s.f.) A1

Marks: M1 for midpoints, M1 for Σfx attempted, M1 for 4180, A1 for 34.8.

Why it is only an estimate. The classes hide the real values, so you place every student at the middle of their interval. Say this on camera — "calculate an estimate" is a deliberate instruction, not padding.

Common mistake: dividing by 5 instead of 120. You divide by the total **frequency**, never by the number of classes. 4180 ÷ 5 = 836 minutes should look absurd and stop you.

Mid-point of 30 < $t \leqslant$ 35 is 32.5, not 32 or 33. Add the two boundaries and halve.

QUESTION 6 CONTINUED

Complete the histogram. [4]

Unequal widths, so the bar height is frequency density: M1

$$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}$$

Class	*Freq*	*Width*	*Height*
$0 < t \leqslant 20$	8	20	**0.4**
$20 < t \leqslant 30$	22	10	**2.2**
$30 < t \leqslant 35$	30	5	**6**
$35 < t \leqslant 40$	34	5	**6.8**
$40 < t \leqslant 60$	26	20	**1.3**

Draw the last three bars at heights 6, 6.8 and 1.3. A3

Marks: M1 for using frequency ÷ width, then A1 for each of the three remaining bars drawn at the right height and width.

The whole question in one sentence: on a histogram, **area** is frequency, not height. Plotting 30, 34 and 26 as heights makes the 5-minute classes look four times too big and scores zero.

Use the two given bars as your proof. The 0–20 bar is drawn at 0.4 and 8 ÷ 20 = 0.4 ✓. The 20–30 bar is at 2.2 and 22 ÷ 10 = 2.2 ✓. They have handed you the method — point this out on camera.

Check the total area: 20(0.4) + 10(2.2) + 5(6) + 5(6.8) + 20(1.3) = 8 + 22 + 30 + 34 + 26 = 120 ✓, the number of students.

QUESTION 6 CONTINUED

Complete the cumulative frequency table and draw the diagram. [2]

Running total of the frequencies: 8, 8+22, 30+30, 60+34, 94+26 B1

$t \leqslant 20$	$t \leqslant 30$	$t \leqslant 35$	$t \leqslant 40$	$t \leqslant 60$
8	30	60	94	120

Plot at the **upper** boundary of each class — (20, 8), (30, 30), (35, 60), (40, 94), (60, 120) — and join with a smooth curve from (0, 0). B1

Use your diagram to estimate the interquartile range. [2]

Lower quartile at $\frac{120}{4}$ = 30th value $\rightarrow$ $t \approx 30$ M1

Upper quartile at $\frac{3 \times 120}{4}$ = 90th value $\rightarrow$ $t \approx 39.4$

IQR = 39.4 − 30

≈ 9.4 minutes A1

Marks: B1 for the table, B1 for the curve. Then M1 for reading at the 30th and 90th values, A1 for an IQR of about 9–10 (your own curve is followed through).

Quartiles on a CF curve use n, not $n + 1$. With 120 students, read across at 30 and at 90. The n+1 rule is for a plain list of data, not for a cumulative frequency graph.

Plot at the end of the class, never the middle. The point is (20, 8), because 8 students took **up to** 20 minutes. Plotting at 10 shifts the whole curve and every reading taken from it.

Draw both quartile lines on the graph. The examiner wants to see the horizontal line at 30 and at 90 and the drop to the axis — it is how the M mark is evidenced.

Question 7

Without replacement

A bag holds 7 red and 5 blue counters. Two are taken without replacement. Complete the tree diagram. [2]

First counter: 12 in the bag — R is $\frac{7}{12}$, B is $\frac{5}{12}$ **B1**

Second counter: only 11 left, and one colour is one fewer. **B1**

After R: R = $\frac{6}{11}$, B = $\frac{5}{11}$

After B: R = $\frac{7}{11}$, B = $\frac{4}{11}$

Find the probability that both counters are red. [2]

Follow the top branch and **multiply** along it:

$\frac{7}{12} \times \frac{6}{11} = \frac{42}{132}$ **M1**

$\frac{7}{22}$ **A1**

Marks: B1 for the first pair of branches, B1 for all four second branches. Then M1 for multiplying along a branch, A1 for $\frac{7}{22}$ (or 0.318).

Along the branch you multiply; between branches you add. That one sentence answers every tree-diagram question on the paper.

Without replacement changes both numbers. The denominator drops 12 → 11, *and* the numerator of the colour you took drops by one. Changing only the denominator is the usual slip.

Check each pair sums to 1: $\frac{6}{11} + \frac{5}{11} = 1$ ✓ and $\frac{7}{11} + \frac{4}{11} = 1$ ✓.

QUESTION 7 CONTINUED

Find the probability that the two counters are different colours. [3]

Two ways for this to happen — red then blue, or blue then red.

RB: $\frac{7}{12} \times \frac{5}{11} = \frac{35}{132}$ M1

BR: $\frac{5}{12} \times \frac{7}{11} = \frac{35}{132}$ M1

Add them: $\frac{35}{132} + \frac{35}{132} = \frac{70}{132}$

$\frac{35}{66}$ (= 0.530 to 3 s.f.) A1

Three counters are taken without replacement. Find P(at least one is blue). [3]

"At least one" — do the opposite. The only way to fail is all three red.

P(RRR) = $\frac{7}{12} \times \frac{6}{11} \times \frac{5}{10} = \frac{7}{44}$ M2

P(at least one blue) = $1 - \frac{7}{44}$ M1

$\frac{37}{44}$ (= 0.841 to 3 s.f.) A1

Marks: M1 for either product, M1 for the other, A1 for $\frac{35}{66}$. Then M2 for P(RRR) (M1 if one fraction is wrong), M1 for subtracting from 1, A1 for $\frac{37}{44}$.

Why the complement. Listing "at least one blue" directly means BRR, RBR, RRB, BBR, BRB, RBB, BBB — seven products. The complement is one product and a subtraction. Always take it.

Common mistake: forgetting that *different colours* happens two ways and answering $\frac{35}{132}$. Order matters when you count the paths, even though the final event does not care about order.

Three draws: the denominators go 12, 11, **10**. Students who keep 11 on the third draw lose the M marks.

Question 8

Transformations and vectors

Describe fully the single transformation mapping triangle A onto triangle B. [3]

A has vertices (1, 1), (3, 1), (1, 4); B has (1, −1), (1, −3), (4, −1).

Same size and shape, but turned — so it is a **rotation**. B1

The rule $(x, y) \to (y, -x)$ is a quarter turn clockwise. B1

Both triangles touch the origin, and it is the only point that does not move. B1

Rotation, 90° clockwise, centre (0, 0)

Draw the image of A after enlargement, scale factor −2, centre (0, 1). [3]

From the centre, go to each vertex, then **twice as far the other way**. M1 A2

(1, 1) → (−2, 1)
(3, 1) → (−6, 1)
(1, 4) → (−2, −5)

Marks: B1 rotation, B1 for 90° clockwise (or 270° anticlockwise), B1 for the centre. Then M1 for the method, A2 for the correct triangle (A1 for two correct vertices).

"Describe fully" means all three facts. A rotation needs the word, the angle *with its direction*, and the centre. Miss one and you lose that mark. And if you name two transformations, you score **zero** — the question says *single*.

Negative scale factor = through the centre and out the other side. The image is upside down and on the opposite side of (0, 1). If yours landed on the same side as A, you used +2.

Find the centre by eye: join each vertex to its image and extend. The lines all meet at the centre of enlargement. One quick check that costs nothing.

QUESTION 8 CONTINUED

OA = $\boldsymbol{a}$, **OB** = $\boldsymbol{b}$. M is the midpoint of AB; N is on OB with ON : NB = 3 : 1. Find **AB**, **OM** and **MN**. [5]

(i) To go from A to B, travel back down $\boldsymbol{a}$ then out along $\boldsymbol{b}$:

AB = $-\boldsymbol{a} + \boldsymbol{b}$ B1

$$\mathbf{AB} = \boldsymbol{b} - \boldsymbol{a}$$

(ii) O to A, then half way along AB:

OM = $\boldsymbol{a} + \frac{1}{2}(\boldsymbol{b} - \boldsymbol{a})$ M1

$= \boldsymbol{a} - \frac{1}{2}\boldsymbol{a} + \frac{1}{2}\boldsymbol{b}$

$$\mathbf{OM} = \tfrac{1}{2}(\boldsymbol{a} + \boldsymbol{b})$$ A1

(iii) ON : NB = 3 : 1, so N is $\frac{3}{4}$ of the way along OB:

ON = $\frac{3}{4}\boldsymbol{b}$ M1

MN = **MO** + **ON** = −**OM** + **ON**

$= -\frac{1}{2}\boldsymbol{a} - \frac{1}{2}\boldsymbol{b} + \frac{3}{4}\boldsymbol{b}$

$$\mathbf{MN} = \frac{1}{4}\boldsymbol{b} - \tfrac{1}{2}\boldsymbol{a}$$ A1

Marks: B1 for **AB**. M1 A1 for **OM**. M1 A1 for **MN**, following through from your own **OM**.

The one rule that solves all of these: to get from X to Y, go backwards to O and forwards again — **XY** = −**OX** + **OY**. Every part above is that rule.

A 3 : 1 ratio means quarters, not thirds. There are 3 + 1 = 4 parts, so ON = $\frac{3}{4}$ of OB. Writing $\frac{3}{3}$ is the most common error in vector questions.

Question 9

Four kinds of sequence

Write down the next term and the nth term of each sequence. [8]

Sequence	*Next*	*nth term*	*Why*
7, 11, 15, 19, …	**23**	$4n + 3$	Linear — first differences all +4
2, 6, 12, 20, 30, …	**42**	$n^2 + n$	Quadratic — second differences all +2
3, 6, 12, 24, …	**48**	$3 \times 2^{n-1}$	Geometric — each term × 2
1, 8, 27, 64, …	**125**	n^3	Cubic — the cube numbers

Linear: difference 4, so it starts $4n$. $4(1) = 4$ but the term is 7, so add 3. B2

Quadratic: second difference 2, so the n^2 coefficient is $2 \div 2 = 1$. Subtract n^2 from each term: 1, 2, 3, 4 — that is n. So $n^2 + n$. B2

Geometric: each term doubles, first term 3, so $3 \times 2^{n-1}$. Check $n = 3$: $3 \times 4 = 12$ ✓ B2

Cubic: 1, 8, 27, 64 are 1^3, 2^3, 3^3, 4^3, so the nth term is n^3. B2

Marks: B1 for each next term and B1 for each nth term, 2 marks per row.

Identify the type before you start. Constant first difference → linear. Constant second difference → quadratic. Constant *ratio* → geometric. Recognising the cubes and the squares on sight saves two minutes.

The classic linear error: writing $n + 4$ instead of $4n + 3$. The difference is the **coefficient** of n, not a number you add on.

Always test $n = 1$ and $n = 2$ in your formula before you move on. Both must give the printed terms.

Question 10

Functions and turning points

$f(x) = 3x - 5$, $g(x) = x^2$, $h(x) = 2^x$. Find $f(-3)$, $fg(4)$, $f^{-1}(x)$ and solve $h(x) = 32$. [6]

(a) Replace x with -3: $3(-3) - 5 = -9 - 5$ B1

$$\mathbf{f(-3) = -14}$$

(b) fg means **do g first**, then f.

$g(4) = 4^2 = 16$, then $f(16) = 3(16) - 5 = 48 - 5$ M1

$$\mathbf{fg(4) = 43}$$ A1

(c) Write $y = 3x - 5$ and swap the roles of x and y:

$x = 3y - 5 \rightarrow x + 5 = 3y$ M1

$$\mathbf{f^{-1}(x) = \frac{x + 5}{3}}$$ A1

(d) $2^x = 32$. Write 32 as a power of 2: $32 = 2^5$.

$$\mathbf{x = 5}$$ B1

fg(4) is not f(4) × g(4). It is a chain: the output of g becomes the input of f. Doing f first gives $g(7) = 49$ — a different, wrong answer.

Check an inverse instantly: $f(3) = 4$, so $f^{-1}(4)$ must be 3. And $(4 + 5) \div 3 = 3$ ✓.

QUESTION 10 CONTINUED

$y = x^3 - 6x^2 + 9x + 2$. Find the two turning points and say which is which. [5]

Differentiate — multiply by the power, then drop the power by one:

$\frac{dy}{dx} = 3x^2 - 12x + 9$ M1

Turning points have zero gradient, so set it to 0:

$3(x^2 - 4x + 3) = 0 \rightarrow 3(x - 1)(x - 3) = 0$ M1

$x = 1$ or $x = 3$ A1

Substitute back into the **original** equation for the y values:

$x = 1$: $1 - 6 + 9 + 2 = \mathbf{6}$
$x = 3$: $27 - 54 + 27 + 2 = \mathbf{2}$ A1

(1, 6) maximum · (3, 2) minimum A1

Marks: M1 for differentiating, M1 for setting it to zero and solving, A1 for both x values, A1 for both y values, A1 for correctly identifying maximum and minimum.

How to tell max from min without more calculus. A positive cubic rises, turns down, then rises again — so the **first** turning point is the maximum and the second is the minimum. State that reason out loud.

Or test the gradient either side: at $x = 0$, $\frac{dy}{dx} = 9$ (positive); at $x = 2$, it is -3 (negative). Gradient going + → 0 → − is a maximum.

Common mistake: substituting $x = 1$ and 3 back into the *derivative* instead of the original curve. That gives 0 and 0 every time — which should tell you something has gone wrong.

Question 11

Speed–time graph and decay

Calculate the acceleration during the first 12 seconds. [1]

Acceleration is the **gradient** of a speed–time graph:

$$\frac{\text{change in speed}}{\text{time taken}} = \frac{18 - 0}{12}$$

1.5 m/s² B1

Calculate the total distance travelled. [2]

Distance is the **area under** the graph. The shape is a trapezium.

Parallel sides: the top (42 − 12 = 30 s) and the bottom (60 s). Height: 18 m/s.

½(30 + 60) × 18 = ½ × 90 × 18 M1

810 m A1

Marks: B1 for 1.5. Then M1 for a correct trapezium area (or three correct pieces), A1 for 810.

Or split it into three. Triangle ½(12)(18) = 108, rectangle 30 × 18 = 540, triangle ½(18)(18) = 162. Total 810 ✓. Same marks — use whichever you trust more under pressure.

Do not read the parallel sides off the time axis carelessly. The top side runs from 12 s to 42 s, so it is **30**, not 42. This single misreading is the most common lost mark on travel graphs.

Units tell you the operation. Gradient is m/s per s = m/s^2, so it is acceleration. Area is m/s × s = m, so it is distance. If you ever forget which is which, check the units.

QUESTION 11 CONTINUED

Calculate the average speed for the whole journey. [1]

$$\frac{\text{total distance}}{\text{total time}} = \frac{810}{60}$$

13.5 m/s B1

A machine worth $24 000 loses 12% of its value each year. Find its value after 5 years. [2]

Losing 12% leaves 88%, so the multiplier is **0.88**. M1

$24000 \times 0.88^5 = 24000 \times 0.527731\ldots$

$12 665.57 (≈ $12 700 to 3 s.f.) A1

Find the number of complete years before its value first falls below $8000. [2]

Try successive powers until you drop below 8000: M1

8 years: $24000 \times 0.88^8 = \$8631$
9 years: $24000 \times 0.88^9 = \$7595$

9 years A1

Marks: B1 for 13.5. M1 for the 0.88 multiplier, A1 for the value. M1 for a correct trial either side of 8000, A1 for 9.

Trial and improvement is the expected method and earns full marks. If you know logs: $n >$ log(8000/24000) ÷ log 0.88 = 8.59, so 9 whole years.

Show the year *before* as well. Writing only the 9-year value does not prove 8 years was still above $8000. Both lines, or no M mark.

Question 12

Circle theorems and similarity

AC is a diameter, angle BAC = 34°, angle CAD = 28°. Find angle ABC, angle BCA and angle BCD, with reasons. [5]

(i) B sits on the circle and AC is a diameter, so ABC is the angle in a semicircle. B1

Angle ABC = 90° B1

Reason: the angle in a semicircle is a right angle.

(ii) Angles in triangle ABC add to 180°:

180 − 90 − 34

Angle BCA = 56° B1

(iii) First find the whole angle at A:

Angle BAD = 34° + 28° = 62° M1

ABCD is a cyclic quadrilateral, so A and C are opposite:

180 − 62

Angle BCD = 118° A1

Reason: opposite angles of a cyclic quadrilateral add to 180°.

Marks: B1 angle + B1 reason for (i); B1 for (ii); M1 for angle BAD and A1 for 118° in (iii).

No reason, no mark. Name the theorem in full. “Angle in a semicircle” scores; “because AC is a diameter” alone does not; “it looks like 90°” certainly does not.

Use the whole angle at A. Part (iii) needs BAD, not BAC. The two given angles sit side by side, so add them before you use the cyclic quadrilateral rule.

QUESTION 12 CONTINUED

The interior angle of a regular polygon is 162°. Find the number of sides. [2]

Go via the exterior angle — it is always the quicker route.

Exterior = 180 − 162 = **18°** M1

Exterior angles of any polygon total 360°:

$n = \frac{360}{18}$

20 sides A1

In triangle PQR, ST is parallel to QR. PS = 6 cm, SQ = 4 cm, ST = 9 cm. Calculate QR. [2]

ST ∥ QR makes triangles PST and PQR **similar** (equal corresponding angles).

Compare whole with whole: PQ = 6 + 4 = **10**, PS = 6

Scale factor = $\frac{10}{6} = \frac{5}{3}$ M1

QR = $9 \times \frac{5}{3}$

QR = 15 cm A1

The trap is PS versus PQ. The scale factor is $\frac{10}{6}$, not $\frac{6}{4}$. Using 6 : 4 gives 13.5 cm and loses both marks — you must compare the **full** side of the big triangle with the full side of the small one.

Sense check: QR must be longer than ST = 9, and PQ is $1\frac{2}{3}$ times PS, so 15 is right. An answer below 9 is impossible.